Lecture #8: ~~paralism~~ parallelism across multiple GPUs

Last week: parallelism within a single GPU: reduce memory access via fusion/tiling

this week: parallelism across multiple GPUs: reduce communication across GPUs/nodes via replication/sharding

Generalized hierarchy (from small/fast to big/slow):

=> L1: single ~~no~~ node, single GPU: L1 cache/shared memory

=> single node, single GPU: HBM

=> single node, multi-GPU: NVlink

=> multi-node, multi-GPU: NVswitch